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AI Health and Fitness Application



"Learning Smarter, Growing Stronger"

Introduction



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- ❑ Quality Assurance
- ❑ Database Coordinator



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- ❑ Frontend Developer
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- ❑ Timekeeper/Scribe



Students have so much on their plate!

- Balancing academic workloads and healthy living
- Irregular schedules and long study sessions
- Overwhelmed, unmotivated, and unsure where to start

Mental and physical health are not as prioritized among students due to these limitations.





Fitness Scholars

A health and fitness app that empowers students to take control of their fitness goals, leveraging AI to assist in workout selections and scheduling.



Team Charter

Our team completed a Team Charter in effort to establish a strong foundation for the project to be built with.

- Team vision and goals
- Roles within the team
- Ground rules
- Plan for gathering team and self-assessment data

This document was signed by all members of the team in agreement that we would abide by what is within the Team Charter





Vision and Scope

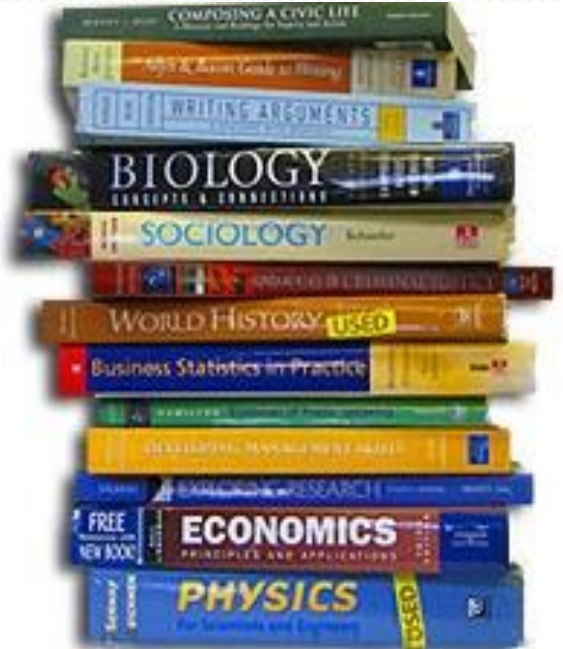
- Our vision is to empower scholars to live a healthier more balanced life through personalized, AI driven fitness recommendations
- We are committed to encouraging long-term wellness, not just short-term results.
- Our app will encourage students to celebrate progress, build confidence, and develop habits that extend far beyond the classroom.





Requirements Document

- We strived to create an app for students with busy schedules so they can stay on track with their studies and their health.
- Using AI we will curate a workout schedule specifically designed for a student's needs.





Core Features

Functional Requirements:

- User Account Login,
- Generating workout based on set schedules
- Curated workout splits to meet User Goals
- AI Chatbot functionality
- Motivational Greeting message,
- Logging completed workouts
- Workout Streaks
- Configurable User Settings

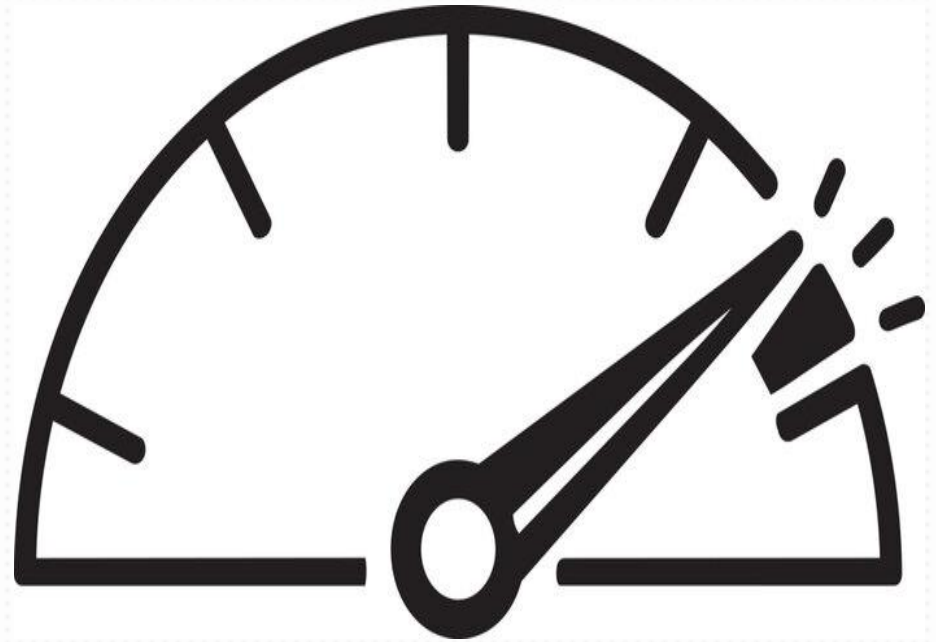




System Quality Requirements

Non-Functional Requirements

- User Friendly UI,
- Fast Performance
- Optimal Load Times
- Reliability
- Mobile Functionality
- Security of User Data





Design Document

- **Architecture:**

- Layered System: Frontend --> Backend --> Database --> AI engine

- **Core Components:**

- Frontend: User interface and dashboards
- Backend: Python logic and processing
- Database: Stores user and workout data
- AI engine: Personalized workout generation

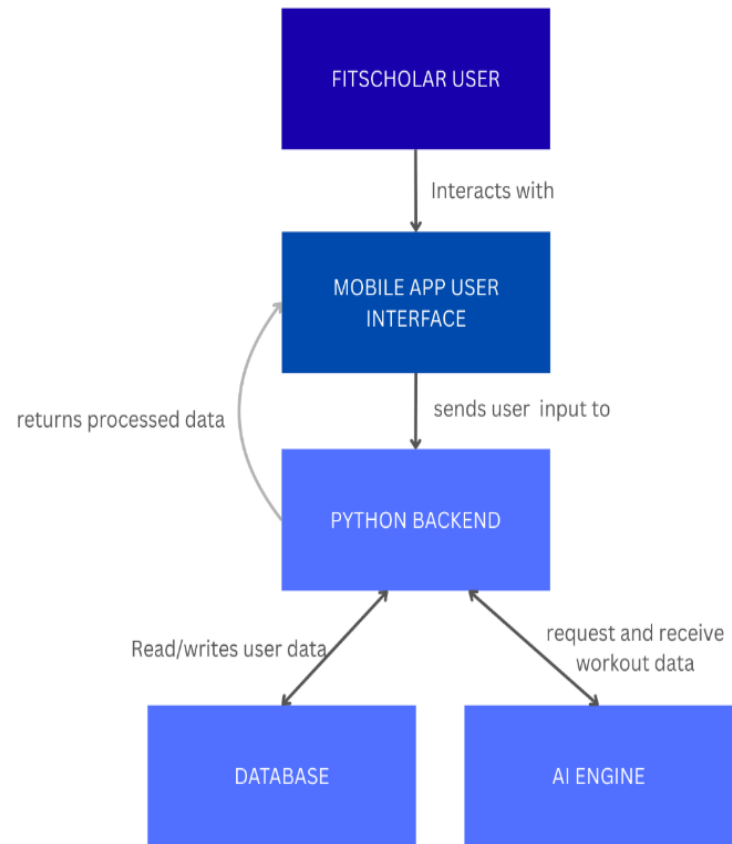
- **Backend Workflow:**

- Validate Input: Check data completeness and format
- Parse Request: Extract user goals and preferences
- Process Logic : Apply rules based on fitness level and goals
- Database Access: Fetch/store user data
- AI Integration: Generate personalized workout plan
- Format Response: Convert results into structured output
- Return to Frontend: Send final data to UI

- **Data Flow:**

- User input --> Backend processing
- AI generates workouts or data fetched from database
- Results returned to UI

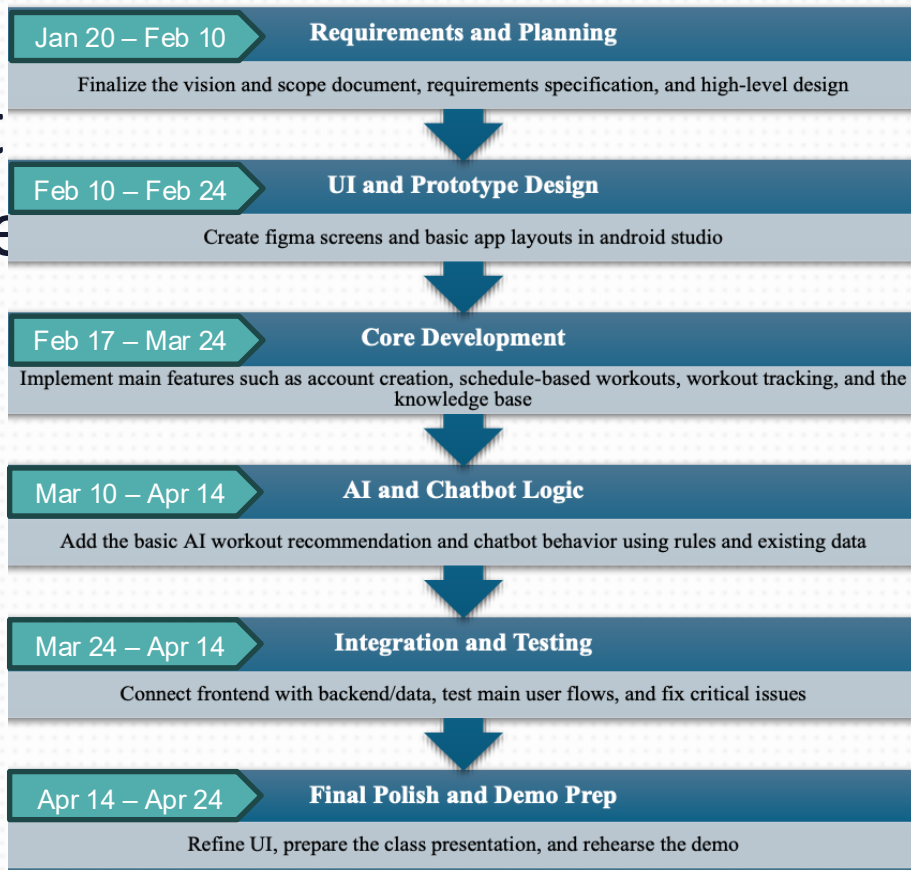
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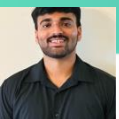




Project Plan Document with Timeline/Schedule

- Broken into **6 phases**: Requirements → Design → Development → AI → Testing → Deployment
- Each phase had specific deliverables
- We also mapped out a **Work Breakdown Structure**, splitting front end, back end, AI, and database across the team
- Identified risks early like time pressure, learning new tools, and integration challenges
- Used **Discord, GitHub, and Zoom** to stay coordinated as a team throughout the semester





Test Strategy

- The testing approach for this application focuses on verifying that each feature works as intended and meets the functional requirements. Testing will be performed manually using an Android emulator and will complete the following:
 - Functional Testing
 - Authentication Testing
 - Schedule Feature Testing
 - Chatbot AI Testing
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Test Plan / Test cases

TC-04: Login with Invalid Credentials

- User clicks “Sign In” and enters incorrect username/password
- System displays error message: “Invalid username/password”
- Message disappears after a short time
- Input fields are cleared

TC-08: Chatbot Answers Workout Questions

- User opens chatbot and asks a workout-related question
- Chatbot processes and understands the input
- A helpful and relevant response is displayed
- Response is given quickly without errors

TC-05: Track Workout Activity

- User selects workout tracking from the home screen
- Chooses a workout activity to log
- Enters details like exercise name, time, sets, and calories
- Clicks to save the workout



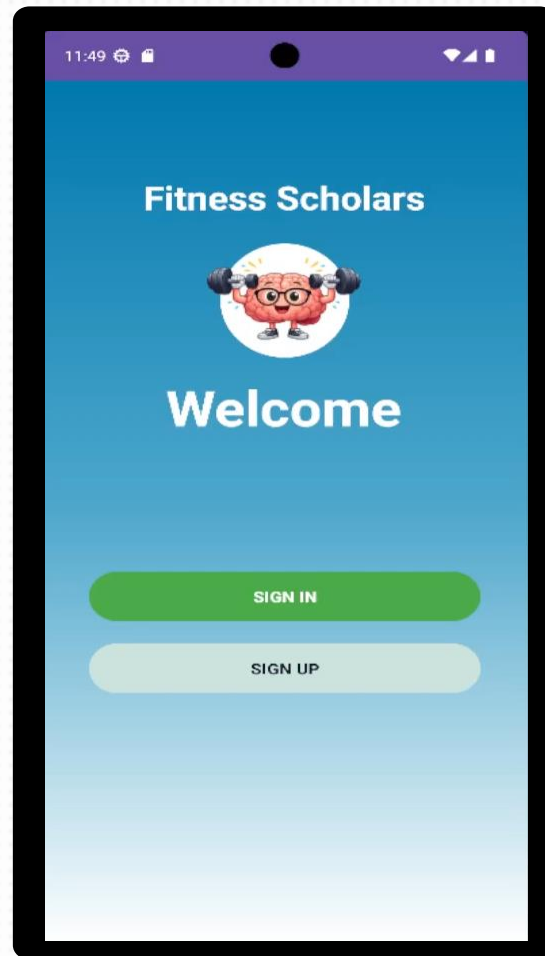


Application Demo



SIGN UP PROCESS

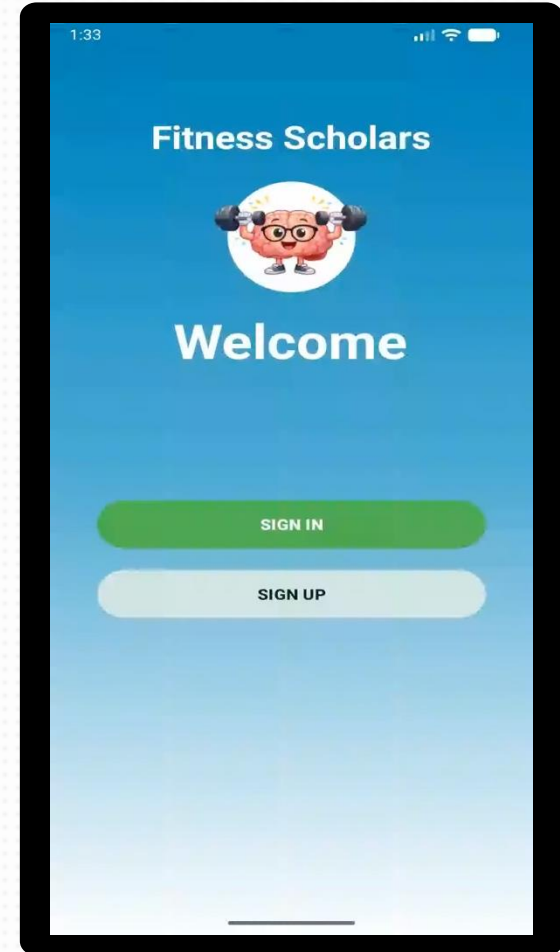
- To create an account:
 - Enter a unique username
 - Pick a password
 - Confirm password
- Errors if username is already in the database and redirects to main screen.





SIGN IN PROCESS

- If a user account already exists in the database, you can use the sign in option to see your workouts.
- To sign into an account
 - Enter your username
 - Enter your password





Entering User

- ## Details
- User enters their details and adjusts their difficulty level
 - The difficulty level is adjusted into 3 sections: Beginner, Intermediate, and Advanced.
 - Then, the user sets their schedule by checking the days they are available and the times they are available.
 - A workout plan is then generated for the user, depending on the difficulty, and timings.

The image shows a mobile application interface for a fitness app. The screen is titled "Tell Us About You" with a subtitle "Enter your fitness details". The form includes the following fields and options:

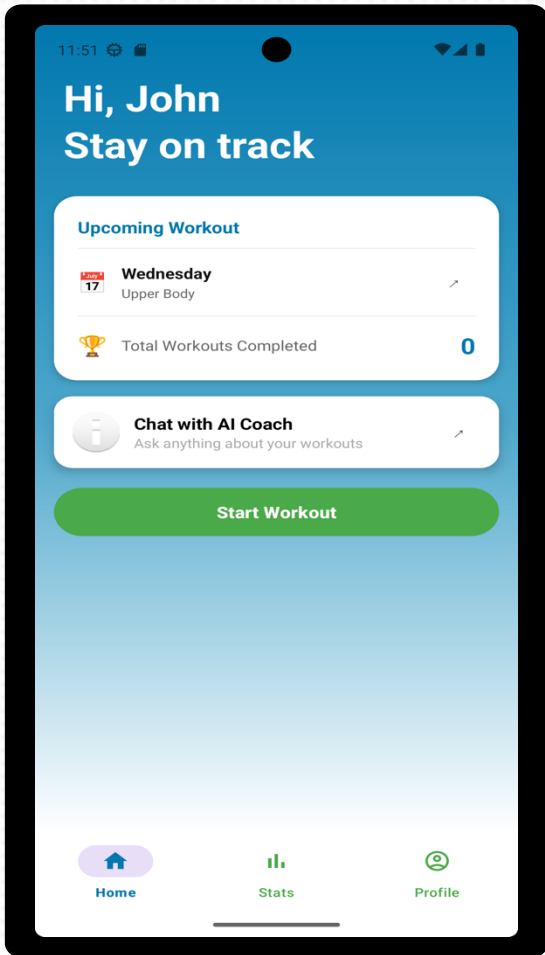
- Age:** A text input field.
- Height:** Two input fields labeled "Feet" and "Inches".
- Weight in lbs:** A text input field.
- Gender, male or female:** A text input field.
- Difficulty:** Three radio button options: "Beginner", "Intermediate", and "Advanced".
- Set My Schedule:** A green button at the bottom of the form.



Home Screen

- The user enters the home page with a welcoming and encouraging message.
- The user can see the day of the week along with the type of exercise routine they have for the day.
- The user can see how many workouts they have completed throughout the day.
- The user has accessibility to our AI chatbot/coach where they ask for workout recommendations.

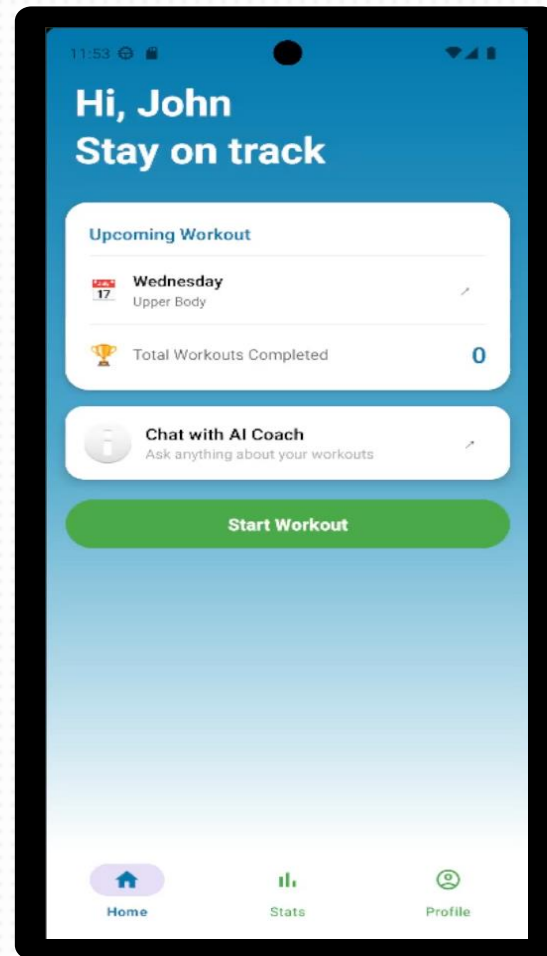
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AI Chatbot

- The user could ask for workout recommendation to their likeness.
- The user can also ask recovery recommendations in case they are feeling fatigue after a workout.
- The user can also use the AI chatbot to generate a new workout schedule.
- The AI chatbot can answer all fitness and workout related questions.
- The AI chatbot does not answer anything of the workout scope.

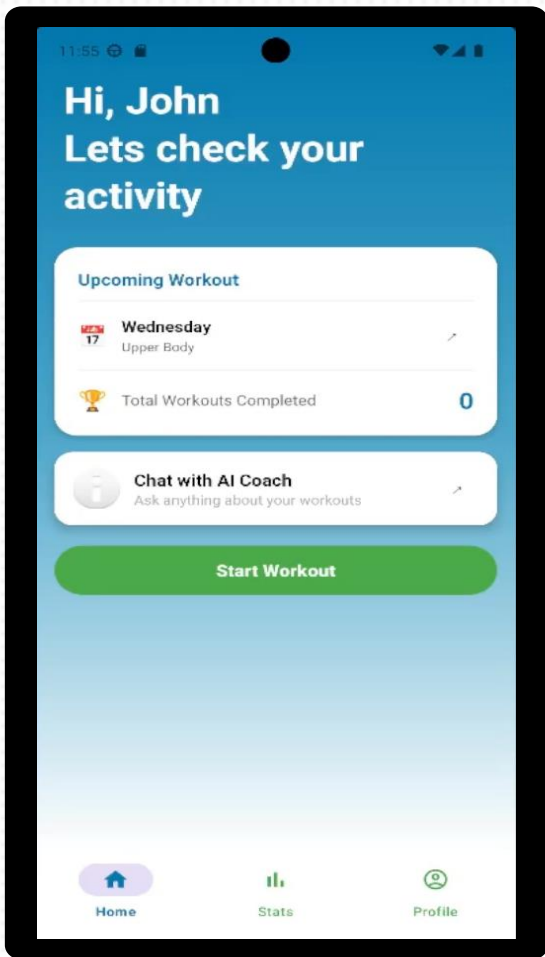




Workout Plan

- Home screen shows your upcoming workout when you open the app.
- Tap **Start Workout** to load a plan based on your schedule and fitness level.
- Browse any day of the week to view planned workouts like Upper Body, Lower Body, or Cardio.
- **Refresh Plan** lets the user generate an updated workout plan.
- Each exercise shows the name, difficulty, target muscle group, sets, and duration.
- After finishing, tap **Workout Done** to save the session.
- The workout is logged automatically, streaks update, and the Stats page reflects progress.

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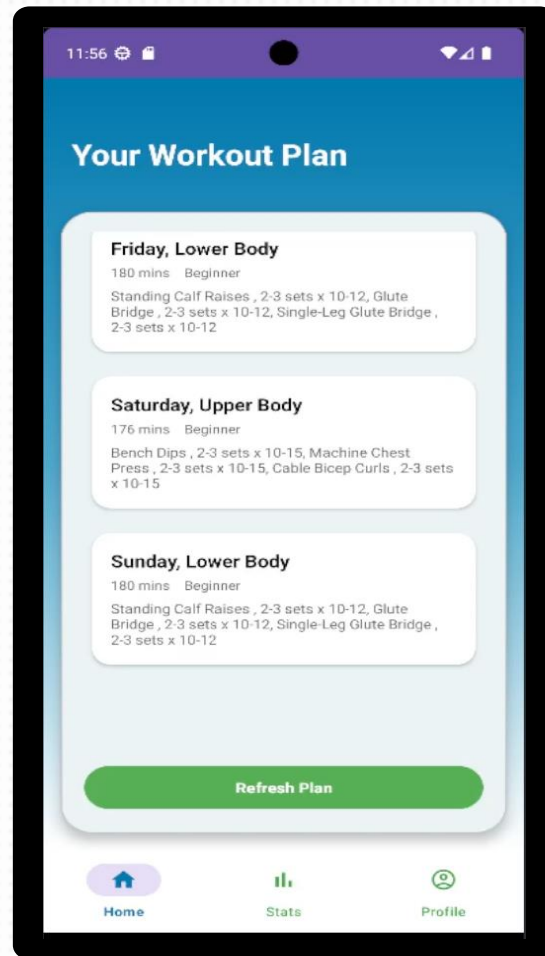


Stats Page

- **Stats Page – Overview & Features**
- User navigates to the Stats Page to track fitness progress
- Displays key stats like total workouts, day streak, and last active date
- Weekly activity calendar shows scheduled (blue) and completed (green) workouts
- Water intake tracker shows daily goal and progress bar
- Recent workouts section shows last completed activity
- Workout plan lists exercises by day with duration and difficulty
- User can set availability and update schedule

Key Features - User Stats Summary , Weekly Activity Tracker, Water Intake Tracking , Recent Workouts, Workout Plan section , and Schedule Management

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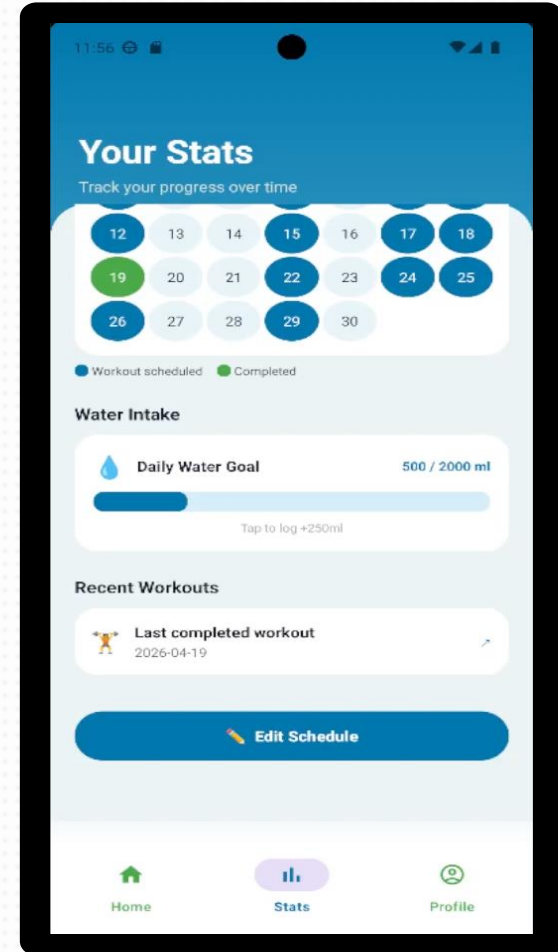
Profile Tab &

Using the navigation bar a User can access their Profile settings at any time.

Settings

In these settings the User can configure options such as:

- Change Username
- Change Password
- Notifications (On/Off)
- Dark Mode (On/Off)
- Delete Account
- Log Out





Lessons Learned & Future

Improvements

- Backend was the biggest challenge, required collaboration across 4 team members to get it working
- Deploying to Render was tricky, server kept crashing, pathing issues, code not being recognized
- Managing so many tools at once was tough, Android Studio, VS Code, Render, OpenAI API, GitHub all running together
- Connecting frontend to backend took more time than expected

Future improvement:

- Add a workout timer built into each exercise
- Better workout UI to guide users through exercises step by step

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THANK

Do you have any questions?

YOU!

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